JAVA LAB – THREADS

Question: Write a Java program that creates threads by extending Thread class. The first thread displays “Good Morning” every 1 second, the second displays “Hello” every 2 seconds and the third displays “Welcome” every 3 seconds. Repeat the same by implementing Runnable.

1. Aim

To create and execute multiple threads using the Thread class and Runnable interface, with different time intervals.

2. Program using Thread class

```
class GoodMorning extends Thread {
    public void run() {
        try {
            while (true) {
                System.out.println("Good Morning");
                Thread.sleep(1000);
            }
        } catch (InterruptedException e) {
            System.out.println(e);
        }
    }
}

class Hello extends Thread {
    public void run() {
        try {
            while (true) {
                System.out.println("Hello");
                Thread.sleep(2000);
            }
        } catch (InterruptedException e) {
            System.out.println(e);
        }
    }
}

class Welcome extends Thread {
    public void run() {
        try {
            while (true) {
                System.out.println("Welcome");
                Thread.sleep(3000);
            }
        } catch (InterruptedException e) {
            System.out.println(e);
        }
    }
}

public class ThreadDemo {
    public static void main(String[] args) {
        new GoodMorning().start();
        new Hello().start();
        new Welcome().start();
    }
}
```

Sample Output

```
Good Morning
Hello
Good Morning
Welcome
Good Morning
Hello
Good Morning
Welcome
Hello
```

Good Morning
(continues repeatedly at the specified intervals)

3. Program using Runnable interface

```
class GoodMorning implements Runnable {
    public void run() {
        try {
            while (true) {
                System.out.println("Good Morning");
                Thread.sleep(1000);
            }
        } catch (InterruptedException e) {
            System.out.println(e);
        }
    }
}

class Hello implements Runnable {
    public void run() {
        try {
            while (true) {
                System.out.println("Hello");
                Thread.sleep(2000);
            }
        } catch (InterruptedException e) {
            System.out.println(e);
        }
    }
}

class Welcome implements Runnable {
    public void run() {
        try {
            while (true) {
                System.out.println("Welcome");
                Thread.sleep(3000);
            }
        } catch (InterruptedException e) {
            System.out.println(e);
        }
    }
}

public class RunnableDemo {
    public static void main(String[] args) {
        new Thread(new GoodMorning()).start();
        new Thread(new Hello()).start();
        new Thread(new Welcome()).start();
    }
}
```

Sample Output

```
Good Morning
Hello
Good Morning
Welcome
Good Morning
Hello
Good Morning
Welcome
Hello
Good Morning
(continues repeatedly at the specified intervals)
```

4. Result

Thus, multiple threads were successfully created using both the Thread class and Runnable interface, and the messages were displayed at 1, 2 and 3 second intervals.